

External API Error Resolution Report

1. Issue Overview

An error was observed in the system while interacting with an external API. The issue caused failure in fetching and processing data, leading to disruption in system functionality.

2. Problem Description

The application encountered intermittent failures when making API requests. The responses were either delayed, returned invalid data, or resulted in timeout errors.

3. Root Cause Analysis

After investigation, the issue was identified as:

- API endpoint instability
- Improper error handling in the system
- Timeout configuration too low
- Lack of retry mechanism

4. Steps Taken to Resolve

The following steps were implemented:

- Increased API timeout duration
- Added retry logic for failed requests
- Implemented proper exception handling
- Validated API response structure before processing

5. Code-Level Fix

Example (Pseudo-code):

```
try:
    response = call_api()
    validate(response)
except TimeoutError:
    retry()
except Exception as e:
    log_error(e)
```

6. Testing & Validation

- Tested API calls under different network conditions
- Simulated failures to verify retry logic
- Confirmed stable responses after fixes

7. Result

The system now handles API failures gracefully. The success rate of API calls improved significantly, and no critical failures were observed after deployment.

8. Future Improvements

- Implement circuit breaker pattern
- Monitor API performance using logging tools
- Introduce fallback mechanisms